

2 The photograph shows switchgrass, *Panicum virgatum*.



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This plant is grown on many farms in North America. It produces a large biomass that can be used to generate electricity or food for farm animals.

A scientist investigated the biomass produced by two different varieties of switchgrass, Almo and Kanlow.

One field was divided into twenty plots of the same area.

Ten of these plots were planted with seeds of the variety Almo and the other ten plots were planted with seeds of the variety Kanlow.

All the plots were planted with the same density of seeds.

All the plots were harvested by machine on the same day.

This was repeated each year for seven years.

The notebook shows the results of this investigation.

Almo biomass $\times 10^3$ kg per hectare	Kanlow biomass $\times 10^3$ kg per hectare
year one 20.1	year one 15.5
year two 19.2	year two 22.2
year three 14.2	year three 14.4
year four 11.4	year four 11.4
year five 11.8	year five 12.6
year six 13.6	year six 15.2
year seven 14.0	year seven 16.5

(a) Write a suitable null hypothesis for this investigation.

(1)

(b) (i) State the meaning of the term **biomass**.

(1)

(ii) When burnt, switchgrass biomass releases 14.25 MJ kg^{-1} of energy.

Calculate the energy released from burning all the Kanlow biomass produced in year four.

Give your answer to **two** significant figures, using standard form.

(2)

..... MJ

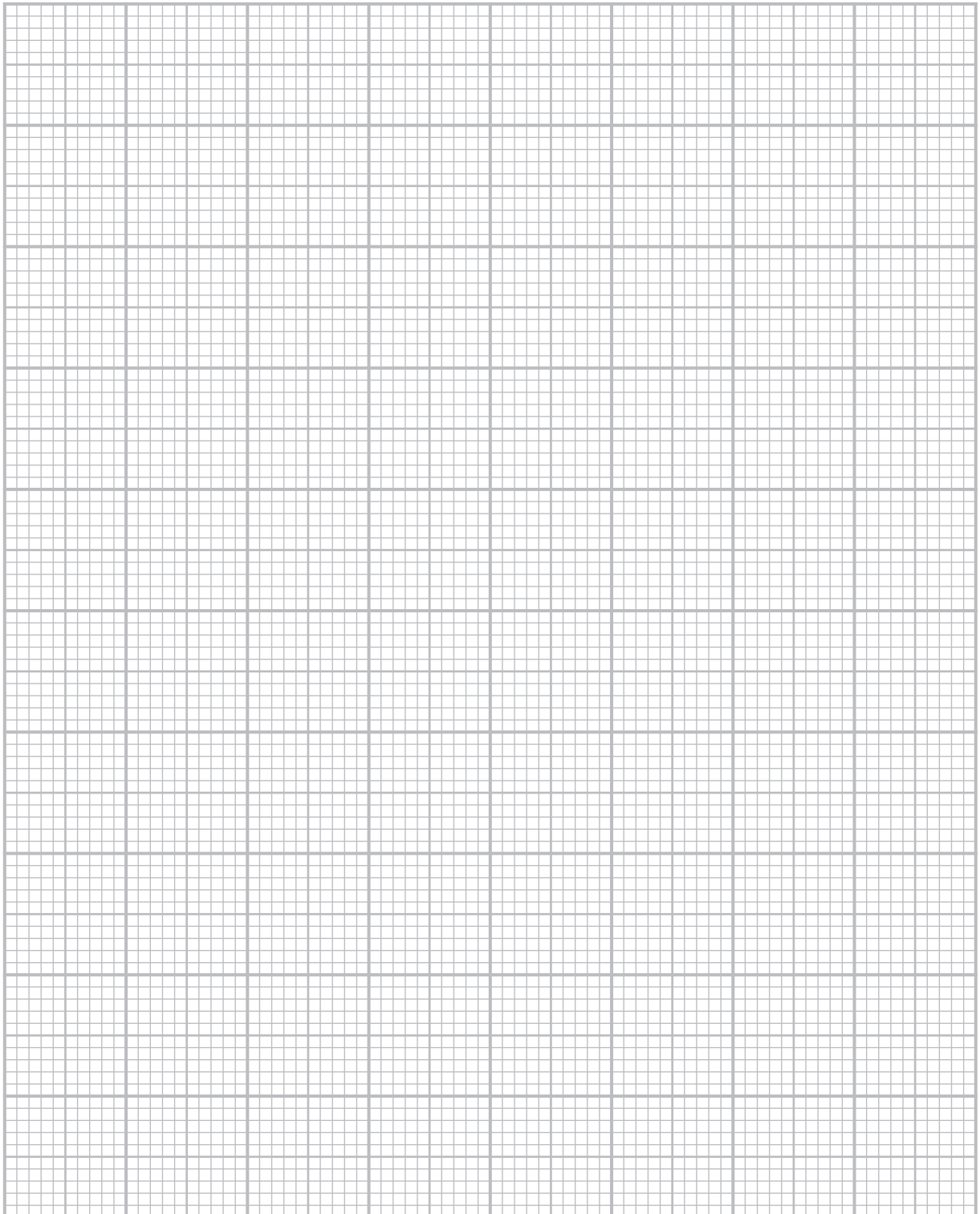
(iii) Draw a suitable table to display the results of this investigation. Label the median for both varieties of switchgrass.

(2)

(c) Draw a graph to show the **median** biomass for each variety of switch grass.

Include an indication of the variability of the data.

(3)



(d) The scientist used a Mann-Whitney U test to analyse the data.

The following formulae were used.

$$U_1 = R_1 - \frac{n_1(n_1 + 1)}{2}$$

$$U_2 = R_2 - \frac{n_2(n_2 + 1)}{2}$$

Where n_1 and n_2 are the number of samples of each switchgrass.

The scientist calculated the value of $R_1 = 57.5$ and $R_2 = 47.5$

The value of $U_1 = 29.5$

(i) Calculate the value of U_2 .

(2)

Answer

For the difference to be significant, the smallest calculated U value has to be equal to, or less than, the critical value shown in the table.

		n_2							
		5	6	7	8	9	10	14	15
n_1	5	2	3	5	6	7	8	13	14
	6	3	5	6	8	10	11	17	19
	7	5	6	8	10	12	14	22	24
	8	6	8	10	13	15	17	26	29
	9	7	10	12	15	17	21	31	34
	10	8	11	14	17	20	23	36	39
	14	13	17	22	26	31	36	55	59
	15	14	19	24	29	34	39	59	64

Use your graph and the information in the table to support your answer.

(3)

[illegible]

(e) Explain why it may **not** be reasonable to draw a valid conclusion from the method used in this investigation and the results obtained.

(3)

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(Total for Question 2 = 17 marks)
